

Polypropylene-nanoclay Composites under Electron Irradiation in SEM: Structure, Charge Trapping and Electron Emission Properties

L. Ben Ammar, S. Fakhfakh

LaMaCop, Université de Sfax
Faculté des sciences Route Soukra Km 3, BP 1171
C.P 3000 Sfax, Tunisie

O. Jbara, A. Hadjadj and S. Rondot

LISM, Université de Reims
UFR Sciences, BP 1039 Moulin de la Housse
51687 Reims Cedex 2, France

ABSTRACT

In this work, polypropylene (PP) and its polymer nanocomposites (PNCs) films containing very low contents (2-6 wt %) of Cloisite 20A natural montmorillonite clay platelets were investigated. In a first, they were characterized by means of several analytical techniques such as X-ray Diffraction (XRD), Fourier Transform Infrared spectroscopy (FTIR), Differential Scanning Calorimetry (DSC) and UV-visible spectroscopy. Secondly, the nanocomposites are submitted to an electron beam in a Scanning Electron Microscope (SEM) and the induced currents are measured using a time-resolved current method. These currents are used to determine the trapped charge and the electron emission yield. The aim is to contribute to the understanding of the effect of different concentrations of nanoclay platelets on the studied properties by correlating the results obtained by both techniques. The analytical characterization showed that upon the increase of nanoclays concentration an intercalated structure and an increase of crystallinity are induced. On the other hand, the optical band gap energy is modified because the increasing of the density of localized states in the forbidden energy band. As regards the measurement of the trapped charge, it was found, surprisingly, that not only the leakage current increases as a function of clay loading level but also the trapped charge (or the surface potential). This could be related to the increase of conductivity in one hand and to proliferation of interfaces between nanoparticles and neighboring materials on the other hand.

Index Terms - Nanocomposites, intercalated structure, displacement current, leakage current, trapped charge, surface potential, electron emission yield.

1 INTRODUCTION

THE automotive, aeronautics and space are looking for materials more efficient, multifunctional, having several characteristics (mechanical, heat resistance, electrical ...) and leading to energy applications. This now leads to the development of new materials and in this case the so-called nanocomposites [1, 2]. The use of nanoscale reinforcements results in better property sets than the compounds taken separately [3, 4]. In the electrical insulation industry in particular, we seek to develop, depending on the intended applications, materials with specific thermo-mechanical characteristics, without neglecting the electrical characteristics.

However, if the insulating composites have the desired thermomechanical properties, the addition of reinforcement is generally accompanied by a decrease in electrical characteristics, such as dielectric breakdown and the occurrence of dielectric losses. Polymer nanocomposites (PNCs) are potentially the solution to these problems because thanks to the combination of the matrix and the reinforcement, new materials with a variety of features can be manufactured. However, it is well established that dielectric properties and charge transport are highly sensitive to microstructure, the distribution of charge, and the defects. Indeed, grain boundaries (i.e. interfaces between different phases) are numerous in nanocomposites and constitute sites for the trapping of electric charges. In addition the influence of these defects on the electrical conductivity of PNCs materials has been shown by several authors [5]. Our goal in this work is to

contribute to the study of the structure of nano composites, especially the dispersion of the nanoparticles in the polymer matrix and levels of intercalation and exfoliation, to understand certain electrical properties of these materials. This study examines polypropylene nanocomposite / clay with very low contents (2-6 wt %) of Cloisite natural montmorillonite clay.

In the first part, structural, thermal and optical properties of PNCs were investigated by XRD, FTIR, DSC, and UV-visible spectroscopy. In the second part charge transport measurements are performed thanks to an instrumented SEM using electrostatic influence phenomenon. The experiments were performed under electron beam irradiation in high vacuum and the charging and discharging processes were observed by time response of current measurements. The characteristics of the trapped charge (amount, time constant), the secondary electron emission are correlated to the structural and optical results.

2 EXPERIMENTAL DETAILS

2.1 MATERIALS AND CHARACTERIZATION TECHNIQUES

All the samples were supplied from the project VAMAS (The Versailles Project on Advanced Materials and Standards) [6]. They are unfilled base PP and its PNCs containing 2 and 6 wt% of clay nanofillers (Southern Clay Products Cloisite 20A), which are called PP0%, PP 2%, and PP6 %. Cloisite 20A (montmorillonite: MMT) is a natural clay phyllosilicate belonging to the smectite family. Its chemical composition is $[\text{Si}_4 \text{O}_{10} \text{Al}^{3+}_{(2-x)} \text{Mg}^{2+x}(\text{OH})_2]$ and is intercalated with dimethyl dihydrogenated tallow. Its nominal interlayer spacing is 2.47 nm [7]. The polymer matrix (PP0%) is a thermoplastic semi-crystalline belonging to the family of polyolefin polymer. Its chemical formula is of the form $-\text{[CH}_2\text{-CHB]}_n\text{-}$, where in B is a methyl group.

The crystalline structures of PP0% and PP nanocomposites PP2%, PP6% was investigated by using X-ray diffractometer (Bruker D8 Advance). The X-ray pattern data was collected from $2\theta = 2^\circ$ to 30° using a monochromatic Cu K α ($\lambda = 1.541 \text{ \AA}$) radiation with a step of 0.06° every 60s. Perkin-Elmer 4000 DSC was used to determine the effect of the nanoclay reinforcement on the thermal properties of the samples; the measurements were performed on 10–12 mg samples under nitrogen atmosphere. The FTIR spectra were obtained in the $4000\text{--}400 \text{ cm}^{-1}$ region using a Nicole Impact 410 FT-IR spectrophotometer. The spectra of various PP and its PNCs PP2% and PP6% were compared in order to determine the occurred changes. The optical property was evaluated using a UV-Visible spectrophotometer (Lambda 950 Perkin Elmer) with a scanning wavelength from 400 to 800 nm, Optical parameters such as the optical band gap E_g and Urbach energy E_u , have been investigated.

2.2 CHARGE TRANSPORT MEASUREMENTS UNDER ELECTRON IRRADIATION

Electric charge transport and charging ability of studied polymers have been investigated thanks to a device based on

the electrostatic influence phenomenon and adapted to a scanning electron microscope (SEM). This device allows measuring simultaneously two induced currents under electron irradiation: the so-called displacement current I_d and the leakage current I_L . A schematic diagram of the experimental arrangement that was described in detail in a previous paper [7] is shown in Figure 1. The investigated PNCs samples for both performed techniques are in the form of sheets with dimensions $1 \text{ cm} \times 1 \text{ cm}$ and a thickness of $135 \text{ }\mu\text{m}$. The samples are locally irradiated (beam diameter $< 1 \text{ }\mu\text{m}$) at room temperature in SEM fast scanning mode (50 frames per second) over an area of typically 28 mm^2 . In order to avoid any residual charge, which may result from previous electron irradiation, the beam is switched on automatically once the sample is centered on the specimen holder using an in situ infrared camera.

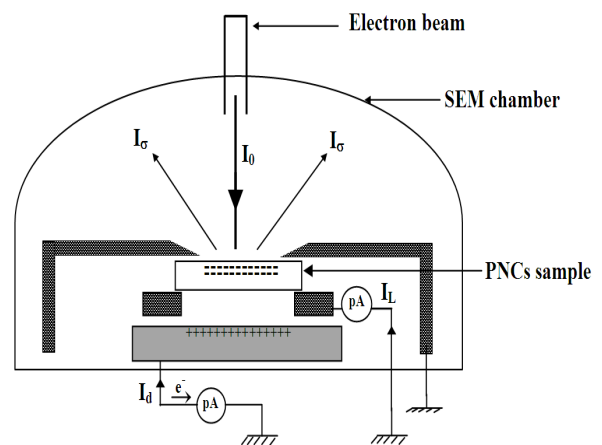


Figure 1. Cutaway view of the experimental arrangement for measuring displacement and leakage currents.

When the surface of the sample is irradiated by a beam of incident electrons with a current intensity I_0 in a SEM, parts of these electrons will be backscattered (BSEs) and others electrons will be emitted by the sample as secondary electrons (SEs) in the vacuum. The BSEs and SEs are combined and termed I_σ , $I_\sigma = \sigma I_0$, where $\sigma = \delta + \eta$ is the total electron emission yield. δ and η are respectively the SE yield and the BSE coefficient. Following the law of conservation of charges, at any instant of time of irradiation, we have:

$$I_0 = \sigma I_0 + I_L + \frac{dQ_t}{dt} \quad (1)$$

I_L is the current that may flow between the sample and ground, resulting from the current through the bulk of the material and that along its surface. $Q_t(t)$ is the net amount of accumulated charges in the sample at time t . Owing to the primary beam energy used, 20 keV, in this study, a negative charge is expected.

This accumulated charge Q_t in the specimen produces a countercharge (i.e. image charge), with opposite sign, Q_{im} , in a grounded metallic electrode isolated from the backside of the sample (Figure 1):

$$Q_{im} = KQ_t \quad (2)$$

Where, K is the electrostatic influence factor that depends on the thickness, dielectric permittivity of the sample, and on the electrical characteristics of the media surrounding the sample. The charge trapping process will produce a displacement current (also called influence current I_d), which corresponds to the induced current on the grounded electrode when electrons are trapped in the specimen:

$$I_d = K \frac{dQ_t}{dt} \quad (3)$$

The time evolution of the accumulated charge deduced by integrating this equation requires knowledge of the K factor (Eq. 3). This factor is calculated using the measured currents during the discharging step as follow:

$$K = -\frac{I_d}{I_L} \quad (4)$$

3 CHARACTERIZATION STUDIES

3.1 MATERIALS AND CHARACTERIZATION TECHNIQUES

Materials were analyzed by XRD to estimate the extent of the nanoclay dispersion. Figure 2a shows XRD spectra of PP polymer and its nanocomposites PP2% and PP6% in the lower angle region, whereas, Figure 2b present changes of crystalline morphology in the large angle region.

Figure 2a shows appearance of peaks in nanocomposites that are absent in the host polymer. Peaks observed between 2° and 3° corresponds to a (001) crystalline plane of nanoclays preserved when compounded with polymeric constituents and the peaks between 6° and 7° can be considered as second order peaks or still the original peak of the montmorillonite before the modification as it was mentioned in the work of Paiva and al [9]. Basal spacing values estimated by using Bragg's law appear on the graph and listed in Table 1. All of these results are consistent with most of the work dealing with this subject in the literature [9]. In PP2% we see a diffraction peak at 2.5° that corresponds to a basal spacing of the (001) plane of 3.53 nm. Compared to a diffraction peak of Cloisite 20A corresponding to 2.47 nm basal spacing of the same plane [7], an increase of 1.06 nm is observed, indicating that an intercalated structure was obtained. PP6% showed peak at 3.13° with basal spacing of 2.82 nm, indicating structures intercalated too (Table 1). Peaks between 6° and 7° were also observed in PP2% and PP6%. The reduction observed in the basal spacing of PP6% can be attributed to aggregation of the particles of the clay during the processing of the materials. This suggests that the higher amounts of the clay made difficult the dispersion of the clay.

Table 1. XRD parameters for PP2%, PP6% and Cloisite 20A.

Nanocomposite	2θ ($^\circ$)	d spacing (nm)	Δ (nm)
Cloisite 20A	3.42	2.47	-
PP2%	2.50	3.53	1.06
PP6%	3.13	2.82	0.35

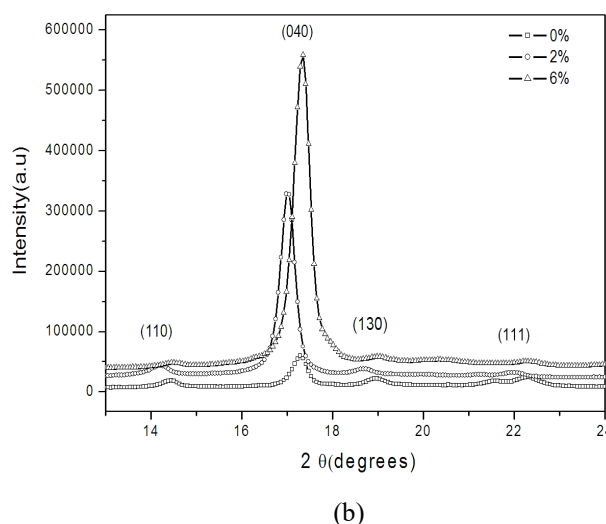
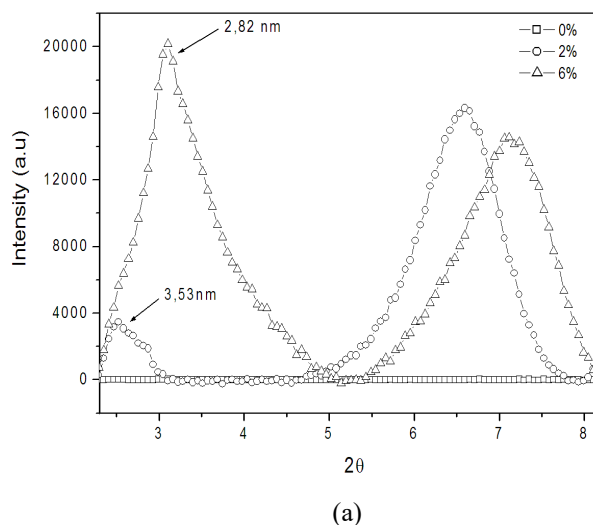


Figure 2. (a): XRD diffraction for neat PP0%, PP2% and PP6% in the lower angle region, (b): XRD diffraction for neat PP0%, PP2% and PP6% at the higher angle region. For a better observation, (040) peaks positions are deliberately shifted

At the higher angle region (Figure 2b), it is observed that five characteristic peaks of most common α -PP structure appear in all the materials at 2θ angles of about 14.0° , 16.8° , 18.6° , 21.2° and 21.9° , corresponding to lattice planes of (110), (040), (130), (111) and (131) respectively [10, 11].

The characteristic diffraction peaks position remained largely unchanged for all the composites, indicating an undisturbed a form crystal structure with the introduction of nanoclay. However, both PP2% and PP6% nanocomposites exhibit a significant increase of the intensity of the crystalline peaks compared to those of PP, indicating that the crystallinity in PNCs increases. This behavior is mainly due to the addition of nanoclay which would promote the nucleation and growth of PP crystallites.

The samples used in this study have already been the subject of several microstructural characterization techniques (X-ray diffraction, TEM, etc) within the frame of the VAMAS project. We therefore refer the reader to the works of UTracki [7] and Nam et al. [12] showing the dispersion of nanoclay

particles in the PP matrix, particularly using TEM, to support our conclusions derived from XRD analysis.

The FTIR spectra of neat PP and its nanocomposites are shown in Figure 3. A pure PP material presents characteristic absorption peaks at 1375 and 1450 cm^{-1} corresponding to C-H bending and a band at 2800-3000 cm^{-1} corresponding to C-H stretching. This stretching band results from the C-H bonds along the aliphatic polymer chains, as well as the side CH_3 groups along these chains. In addition to the specific peaks of PP, as nanoclay is added to the host polymer, characteristic absorptions peaks and bands proper to Cloisite natural montmorillonite appear. A small band at 1020-1050 cm^{-1} that indicates a stretching vibration of Si-O-Si is seen only in PP6% and signifying that a bad dispersion of nanoclay occurs in these PNCs. When the spectrum of PP0% is compared with the spectrum of nanocomposites PP2% and PP6%, small bands at 1,020–1,050 cm^{-1} relate to the stretching vibrations of Si-O-Si from organoclay were seen only in the spectrum of nanocomposites, signifying that the polymer chains have intercalated into the gallery of organoclay [12].

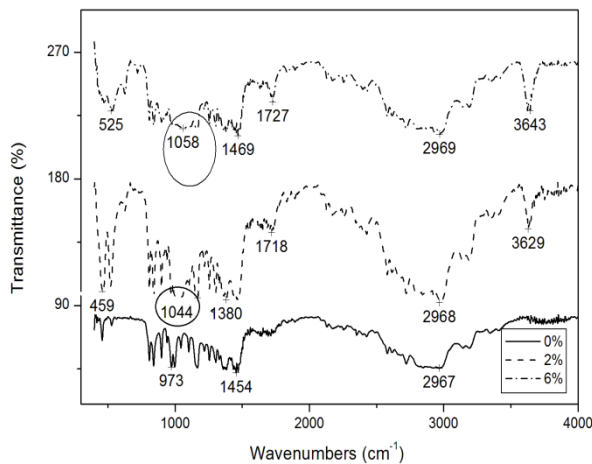


Figure 3. Comparison of FTIR spectra of neat PP and its nanocomposites.

Two peaks at 525 and 460 cm^{-1} correspond to Al-O and Si-O bending vibrations, respectively [13]. A strong absorption band was also observed at 3624 cm^{-1} that corresponds to OH band vibration.

The DSC analysis of PP matrix and its nanocomposites PP2% and PP6% gives curves that are displayed in Figures 4a and 4b.

The corresponding characteristics (T_g , T_c , T_m and χ) are listed in Table 2. We notice that the melting temperature, T_m , of neat PP is 162.75 $^{\circ}\text{C}$ and increases slightly to 163.33 $^{\circ}\text{C}$ by adding 6% of nanoclay to PP matrix (Figure 4a), the T_c increases also slightly with content of nanoclays (Figure 4b). These results are in agreement with some earlier studies that reported no significant changes in T_m values of PP-based nanocomposites using various clays [14-17]. For neat PP, the glass temperature, T_g , is located at -8 $^{\circ}\text{C}$ and increases significantly with nanoclays rate. The degree of crystallinity χ is deduced using the following expression:

$$\chi(\%) = \frac{\Delta H_m}{(1 - W_{\text{nanoclay}}) \Delta H_m^0} \quad (5)$$

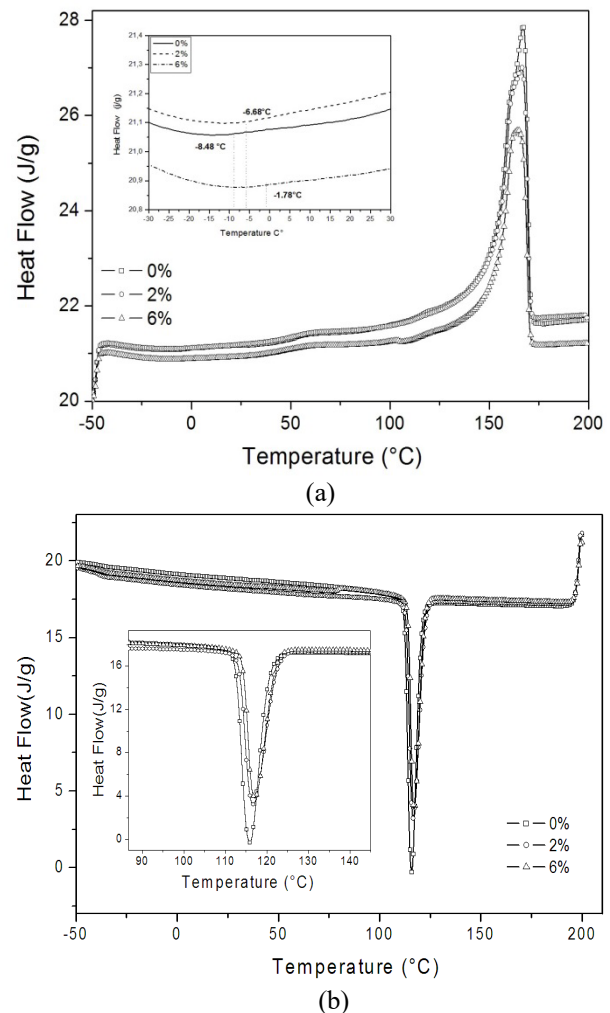


Figure 4. DSC cooling (a) and melting (b) thermograms of PP, PP2% and PP6%. The inset in (a) is an enlarged view around T_g . The inset in (b) gives an enlarged view of T_m .

Where ΔH_m is the fusion enthalpy of the sample, ΔH_m^0 is the fusion enthalpy of crystalline sample and taken in this work as 209 J/g [18], and W_{nanoclay} is the weight fraction of nanoclay in the nanocomposite. The DSC results are summarized in Table 2.

Table 2. DSC results of PP, PP2% and PP6%.

Sample	T_g ($^{\circ}\text{C}$)	T_m ($^{\circ}\text{C}$)	T_c ($^{\circ}\text{C}$)	ΔH_m	χ (%)
PP0%	-8.48	162.72	116.16	87.48	41.85
PP2%	-6.68	163.25	116.92	86.96	42.45
PP6%	-1.78	163.33	117.25	90.74	46.18

From Table 2, it can be noticed that the crystallinity degree χ % of PP increases when the nanoclay particles are added. This trend is in agreement with XRD results. The rise in PP crystallinity suggests that the added clay in the polymer matrix has a nucleation effect. Many works sustain this finding [19].

On the other hand; the increase in glass transition temperature of PNCs over base polymer might suggest that a reduction in polymer chain mobility occurs in the interaction zone contributing to the reduction in polymer chain relaxation. The variation in temperature T_c is related to the crystallization ability of the molecular chains. Reducing the molecular mobility reduces this ability chains to order but promotes the formation of more compact crystalline regions [20, 21].

3.2 OPTICAL CHARACTERIZATION

The optical properties of pure PP and its PNCs (PP2% and PP6%) were studied within the 250 – 850 nm range. The total transmittance of the sample is displayed in Figure 5a. For all samples, the transmittance exceeds 80% to tend to 90% above 350 nm. The red-shift of the transmittance spectra, when the nanoclay content increases, is synonymous of a bandgap narrowing.

The accurate knowledge of the sample thickness allows an easy determination of the absorption coefficient α (Figure 5b) thanks to the OJL model [22].

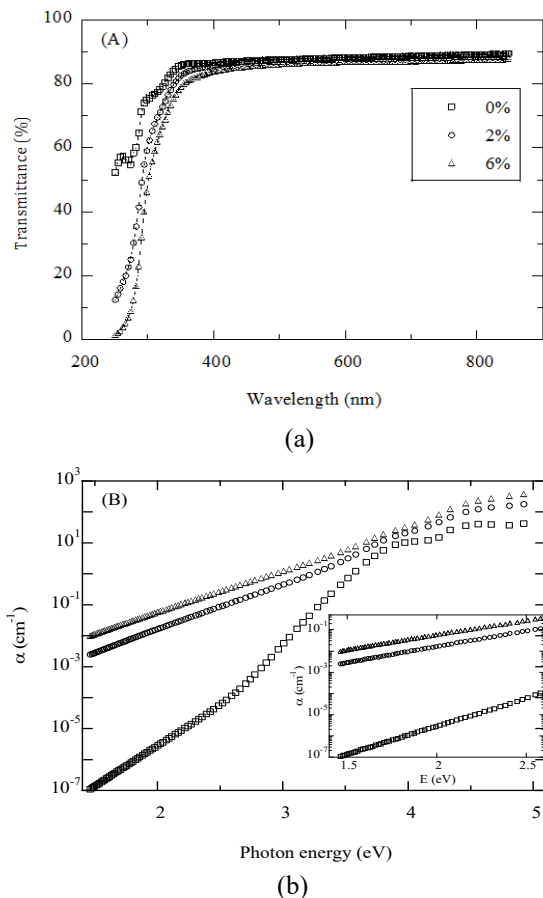


Figure 5. Optical properties of the nanoclay PP composites: (a): Experimental (symbols) and fitted values of the transmission spectra. (b): The absorption coefficient. The inset focuses on the low energy part.

The absorption coefficient α as a function of the photon energy (E) is displayed in Figure 5b. According to this figure, the absorption edge of pure PP is well below 200 nm in agreement with the literature data [23]. An increase in the

nanoclay particles contents up to 6 % red-shifts the absorption edge of the material. It also drastically enhances the film absorption, especially at high photon energy (E).

The inset shows the exponential trend of the low absorption part as a function of photon energy:

$$\alpha(E) = \alpha_0 \exp\left(\frac{E}{E_u}\right) \quad (6)$$

Where the Urbach energy (E_u) is related to the width of the tail of localized states in the band gap which appear because of disorder in the polymer film [24].

The optical bandgap (E_g) can be determined through the Tauc plot [25].

$$\alpha \cdot E = k(E - E_g)^n \quad (7)$$

Where k is a constant and the exponent n depends on the kind of the prevailing optical transitions. For indirect (indirect) allowed transition $n = 1/2$ ($n = 2$), the extrapolation of the straight line of the Tauc plot $(\alpha \cdot E)^n$ onto the energy axis gives the direct (indirect) bandgap E_g .

The determined values of bandgap energy and Urbach energy are reported in Figure 6. In the case of PP, the reported values of E_u can reach some eV [26, 27]. Our values, which are similar to that found by Magda et al [28] and Saqa'n [29], show a linear increase with the nanoclay particles content, due to an enhancement of disorder in the polymer matrix. Simultaneously, E_g sharply decreases at a nanoparticle content of 6%. The value of the bandgap in our pure PP is similar to that reported by some authors [30-32, 23]. The most likely interpretation of such decrease is that nanoclay particles produce disorder and multiple states in the polypropylene valence band structure leading to the narrowing of the bandgap.

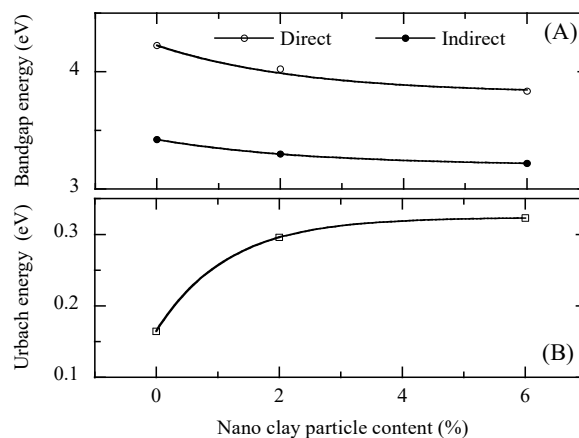


Figure 6. Direct and indirect bandgap energy (A) and Urbach energy (B) as a function of the nanoclay particle content.

3.3 CHARGE TRANSPORT MEASUREMENTS

3.3.1 DISPLACEMENT AND LEAKAGE CURRENTS

Figures 7a and 7b show respectively the time decay of displacement and leakage currents during and after electron beam irradiation for PP and its PNCs (PP2% and PP6%).

These currents were measured simultaneously using the arrangement schematically sketched in Figure 1. The irradiation was performed at 20 keV primary beam energy and 1 nA fixed primary beam current. During the irradiation step the leakage and displacement currents are negative but their shapes are different. For each sample, the displacement current decays rapidly, reaching a very low value after a few tens of seconds and thereafter remains nearly zero. At the beginning of irradiation, the trapping rate is maximal and takes the larger absolute value, I_d^0 for PP6% (Table 3). The leakage current has a minimum value at the beginning of irradiation and increases continuously to finally reach a saturation value I_L^S (Table 3) note that in the case of PP6% the leakage current is significantly higher.

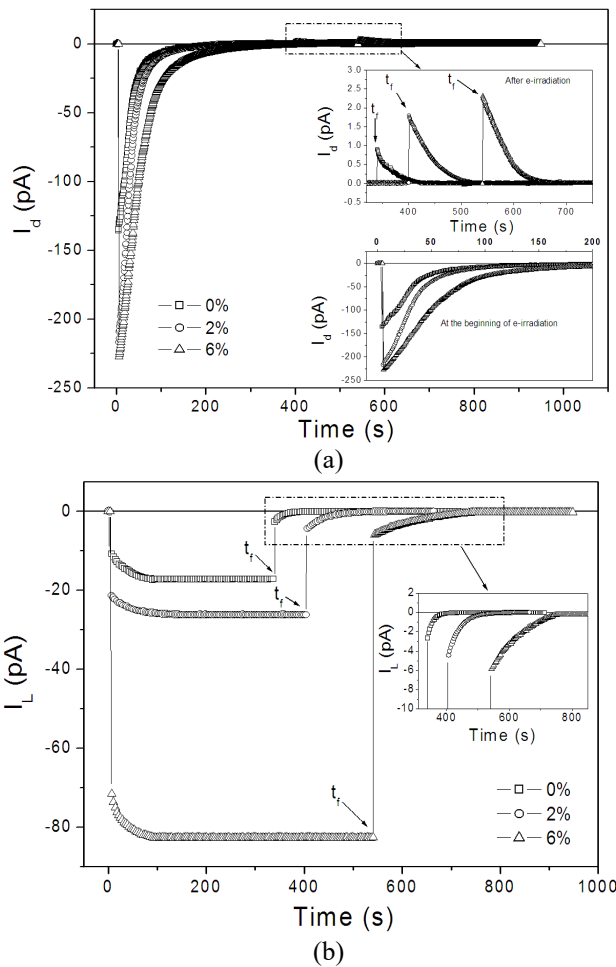


Figure 7. Time variation of displacement current (a) and leakage current (b) during and after electron irradiation of PP and its PNCs. t_f is the instant when the electron beam is blanked.

Table 3. Absolute values of the displacement current at the beginning of irradiation and of the leakage current at the steady state.

Sample	t_f (s)	I_d^0 (pA)	I_L^S (pA)
PP0%	340.47	135.56	17.32
PP2%	400.33	217.12	26.18
PP6%	540.25	227.19	82.62

During the discharge period that begins at time t_f (see the enlarged views in insets of Figures 7a and 7b and the values in Table 3), the displacement current abruptly becomes positive (change in initial sign) and then decays to zero, whereas the leakage current abruptly begins to decrease in magnitude while remaining negative (keeping its initial sign) and then decays also zero. As it is shown on the enlarged views in insets of (Figures 7a and 7b), the magnitude of discharge current depends on the filling rate of nanoclay particles. It is found that the two currents (leakage and displacement) are greater for the highest rate.

3.3.2 TRAPPED CHARGE KINETICS

As it was seen above (Section 2.2), the time integration of the measured displacement current represents the time evolution of the image trapped charge. The trapped charge is deduced from equation (2) with an average value of K equal to 0.32 for PP0%, 0.34 for PP2%, and 0.38 for PP 6% respectively. These values of K were obtained by using the amplified part of decay charge stage (inset of Figures 7a and 7b) following equation (3)). Finally, the measured trapped charges Q_t versus time of electron irradiation of PP and its PNCs are shown in Figures 8a and 8b. With the onset of electron irradiation, the trapped charge increases with time and goes through saturation value Q_t^∞ . The same general behavior of Q_t versus time was observed for all samples but the trapped charge at the steady state increases as nanoclay filling level increases.

Charge trapping is assumed to satisfy first order kinetics as follow:

$$\frac{dQ_t(t)}{dt} = \frac{Q_t^\infty}{\tau} \exp\left(-\frac{t}{\tau}\right) \quad (8)$$

By fitting all of the charge buildup data to this model, the parameter fitting, Q_t^∞ and τ , respectively, the trapped charge at steady state and the time taken for the specimen to fully charge up (time constant) are listed in Table 4. The charging time constant, τ , involves several parameters and in particular the secondary emission, the leakage current, and electron (hole) trapping. Indeed, during irradiation; the charge evacuated via secondary electron or leakage current, is overcompensated by the charge injected. The result of this injection is the increase with time of the amount of mobile charges $\partial Q(mob)/\partial t$ and also the increase of the trapped charges $\partial Q(trap)/\partial t$. When the evacuation of mobile charges is negligible during the irradiation time, the total charge temporal evolution is expected to look similar to the charge of a capacitor but the physical mechanisms are different and involve the change with time of $\delta + \eta$ (Section 2.2) [33]. The steady state trapped charge, Q_t^∞ , may only be attained when any additional charge cannot be trapped.

The results show clearly that Q_t^∞ and τ depend on nanoclay concentration and both increase with increasing this concentration.

When the concentration of nanoclay increases, but does not reach the percolation threshold, the sample is theoretically

able to trap more charges (i.e more traps when clay concentration increases) and takes more time to form a stable space charge Q_t^∞ .

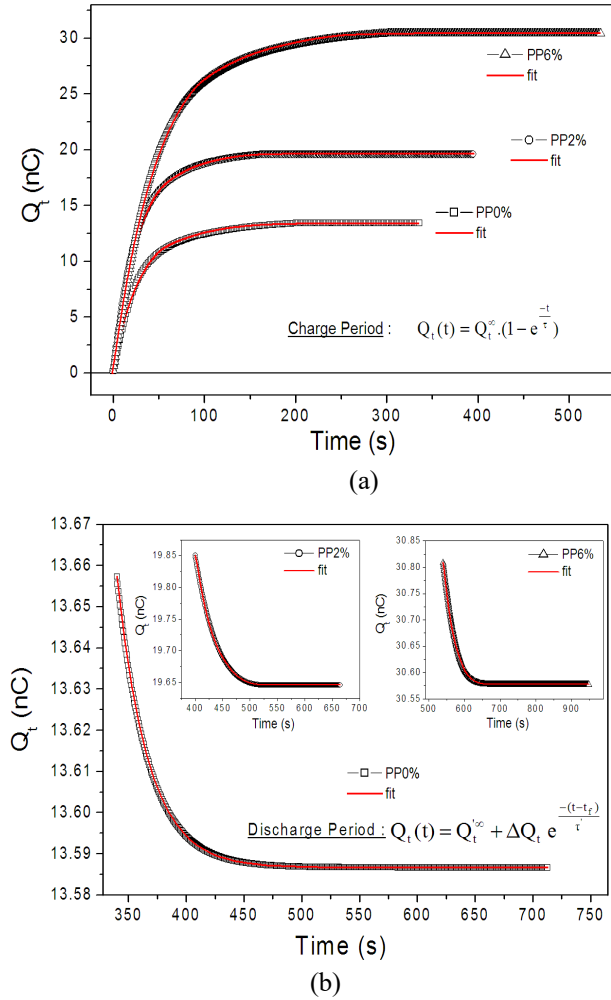


Figure 8. Experimental (symbols) and fitted (curves) trapped charge versus time under (a) and after electron- irradiation (b) of PP and its PNCs at 20 keV primary beam energy and fixed 1 nA primary beam current.

Although, the discharging process is weak (see enlarged view in insets of Figure 8), as for charging the discharging rate after beam blanking ($t > t_f$) obeys also a first order kinetics:

$$\frac{dQ_t(t)}{dt} = \frac{\Delta Q_t^\infty}{\tau'} \exp\left(-\frac{(t - t_f)}{\tau'}\right) \quad (9)$$

ΔQ_t^∞ is the stored charge evacuated from the sample during the time of discharging, $\Delta Q_t = Q_t^\infty - Q_t'^\infty$ where $Q_t'^\infty$ is the charge that remains stored in the sample at the end of discharging step and Q_t^∞ is the stored charge at the steady state of charge (time t_f). τ' and $Q_t'^\infty$ resulting from the fitting to the first order kinetics are given in (Table 4).

To explain the physical meaning of constant time τ' of the discharging process, various time constants may be considered: i) The time constant for secondary electrons to

escape into the vacuum after switching off the beam (low compared to the others time constants). ii) The detrapping time constant related to the trapped charges characterized by their trapping energy (the detrapping process is triggered when a critical value of the internal electric field is reached). iii) The dielectric time constant due to more or less mobile charges such as injected charges before their possible trapping or after their detrapping as well as polarons. iv) The evacuation-to-ground time constant governed by ohmic mechanisms, during which the mobile charges move from the interior of the specimen to its surface (or bulk) and next to ground (It involves the surface conductivity of the specimen or the bulk conductivity especially for thin samples).

The other relevant parameter to describe the charge releasing process is the parameter $\beta = \Delta Q_t / Q_t^\infty$ that quantitatively expresses the variation of the released charge. It is also given in Table 4.

The parameter β depends on the sample properties and represents the stability of stored charge. A dominance of stable trapping occurs when it takes low values while a governing spreading behavior occurs for its high values. One could observe that τ' decreases with the nanoclay content while α increased slightly but remains low.

Table 4. Characteristics of charging and discharging processes.

Sample	PP0%	PP2%	PP6%
$Q_t^\infty \pm \Delta Q_\infty$ (nC)	13.65 ± 0.02	19.85 ± 0.02	30.78 ± 0.02
τ (s)	32.74	38.22	56.92
$Q_t'^\infty \pm \Delta Q_t'^\infty$ (nC)	13.59 ± 0.01	19.70 ± 0.01	30.48 ± 0.01
τ' (s)	52.67	43.47	34.64
β (%)	0.46	0.78	0.95

3.3.3 SURFACE POTENTIAL

Measurement of the surface potential is based on the shift of the high energy cut-off (Duane-Hunt limit) of the x-ray continuous radiation (bremsstrahlung) emitted from the charged sample under electron irradiation at the steady state. The x-ray spectrum was monitored using SiLi energy dispersive spectrometer (EDS). The followed procedure was described in detail in a previous paper [34]. The surface potentials obtained at the steady state for PP and its PNCs are listed in (Table 5). As expected, the surface potential increases when the rate of nanoclay particles increases. This confirms the results obtained above for the trapped charge, because this last one is, at the steady state, directly related to the surface potential.

Table 5. Steady state trapped charge and surface potential at a beam energy $E_0 = 20$ keV and a primary beam current $I_0 = 1$ nA.

Samples	V_s (eV)	Q_t^∞ (nC)
PP0%	9.03	6.67
PP2%	9.50	7.54
PP6%	10.51	11.94

3.3.4 ELECTRON EMISSION

In addition to the charging study, the total electron emission yield during electron bombardment period was determined. The method consists of correlating the measured leakage and displacement currents during this period (Figures 7a and 7b). The total electron emission yield σ defined as the ratio between the total number of electrons emitted by the surface and the number of impinging electrons is deduced from the following charge balance relation:

$$\sigma = \frac{I_{\sigma}}{I_0} = 1 - \frac{I_L}{I_0} - \frac{1}{K} \frac{I_d}{I_0} \quad (10)$$

When the steady state is reached, the displacement current I_d is equal to zero. At that time the total emission yield σ^S is equal to:

$$\sigma^S = \frac{I_{\sigma}}{I_0} = 1 - \frac{I_L}{I_0} \quad (11)$$

Finally, the obtained time evolution of σ for the three studied samples, is shown in Figure. 9.

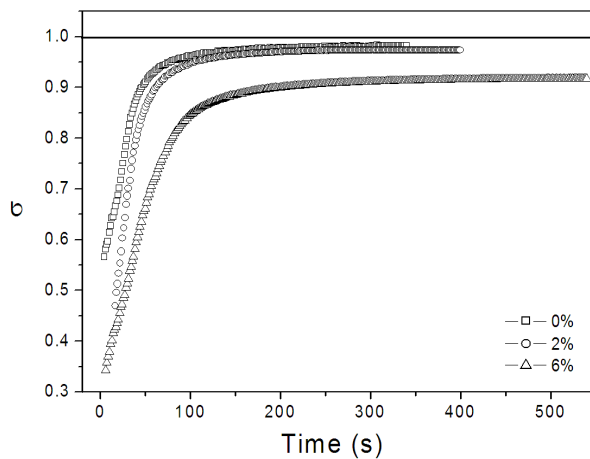


Figure 9. Time variation of total electron emission yield during electron irradiation at 20 keV primary beam energy and 1 nA primary beam current.

As can be seen, regardless of the nature of the sample, σ increases continuously from its value at the early beginning of irradiation, σ_0 , toward a stationary value σ^S (steady state) as the electron irradiation progresses. This time evolution may be easily explained if one considers the slowing down of incident electrons caused by the external electric field induced by the negative charging and the corresponding change of electron emission yield. Indeed, the charging decreases the effective landing energy of incident electrons, and hence increases the electron emission yield [35]. Table 6 summarizes the effect of nanoclay particles level on the values of σ_0 and σ^S obtained at 20 keV and 1 nA. When this level increases both σ_0 and σ^S decrease.

Table 6. Values of σ_0 and σ^S obtained at 20 keV primary beam energy and 1 nA primary beam current for the investigated samples.

Samples	σ_0	σ^S
PP0%	0.566	0.983
PP2%	0.346	0.973
PP6%	0.342	0.917

4 DISCUSSION

According to the results of XRD and FTIR, both PP6% and PP2% have intercalated structures, and therefore a bad nanoclay dispersion occurs. This bad dispersion more pronounced in PP6% favors the formation of particle aggregates in polymer. Moreover, PP6% is the material which presents the highest crystallinity, confirmed by DSC (Table 2), and therefore could be the richest in interfaces. The increase in the sample crystallinity with the nanoclay content has already been reported by P.H. Nam et al [12]. The authors suppose that the clay nanoparticles reduce the PP chain mobility owing to the narrow space surrounded by the dispersed clay particles and the intercalation of the PP chains in silicate galleries. Consequently, the reduction in the PP chains mobility produces smaller and less-ordered crystallites. This growth of interfaces in PNCs has a great influence on charge transport properties, because these interfaces are the seats of a multitude of defects with a high density. Indeed, this higher density of defects leads to increased numbers of localized levels within the forbidden band of the material [28]. In the case of ours PNCs, a well-ordered crystalline structure appears, to which can associate a conduction band and a valence band. The nanoclay reinforcements, acting as localized states, induce a significant modification of their energy diagram and therefore a decreasing of their band energy gap with respect to host polymer. This is confirmed by UV-visible measurement. Taking into account these considerations, one may say that the increase of content of nanoclay in host PP may modify the density of traps and therefore leads to an accentuation of trapping processes under electron irradiation of polymers. It follows that, an increase of the surface potential and of the trapped charge at the steady state. The result of the addition of nanoparticles is a drastic enhancement of the charge storage ability of PNCs. However, the effectiveness of the nanoparticles is such that the amount of material added is normally less than around 5% by weight as it has been verified by numerous studies in the literature [36]. Note that photoluminescence measurements were also used to probe traps induced by the dispersion of nanoparticles in a polymer matrix [37].

The electron emission measurements may also be explained using the same considerations. Indeed, this emission depends on specific crystalline state of the sample and particularly on the interaction of the generated electrons with electrons, phonons, or with defects during their transport toward the surface. As a result the electron emission yield would be large in the case of lack of these interactions and also of defects. Thus, the observed decrease of electron emission is due to the increase in interactions during the transport of electrons toward the surface of sample [35] leading to the reduction of their mean free path because they are scattered and trapped by defects that are more important when the percentage of clay in the PNCs is higher. As a consequence, for a given energy, σ_0 and σ^S decrease as the nanoclay content increases (Table 6).

5 CONCLUSION

From the studies performed using different analysis techniques, a change in structural, thermal, and optical properties is highlighted depending on the amount of the nanoclay that was added in PP. The XRD results revealed a shift in the position of (001) planes meaning an increase in the basal spacing of these planes as the concentration of nanofiller increases. This confirms the occurrence of organic molecule intercalation between nanoclay platelets. However, no significant impact on the PP- α crystalline structure was detected. Both results of XRD and DSC showed that the crystallinity of PNCs increases with the nanoclay concentration. The UV-visible measurements indicate that the band energy gap decreases as a result of adding of nanoclay platelets. This decrease was explained based on the higher density of localized states in forbidden band energy. This finding is then correlated to the trapping charge behavior. Indeed, as the concentration of nanoclay platelets increases, both trapped charge and the surface potential at the steady state, when PNCs are submitted to electron irradiations, increase.

The electron emission yield deduced from the leakage and displacement currents allows also to study the influence of the percentage of clay in the nanocomposite on the charging effects. As expected for a given concentration, the yield increase during irradiation because the surface potential increases, while it decrease when the concentration increases.

This work concern more generally the understanding of charging and discharging, under electron irradiation, of PP nanocomposites resulting from incorporation of nanosized clay particles into a matrix of PP standard material. The result of the addition of nanoparticles is a drastic enhancement of the charge storage ability of PNCs.

REFERENCES

- [1] A. Okada and A. Usuki, "Twenty years of polymer-clay nanocomposites," *Macromol. Mater. Eng.*, Vol. 291, pp. 1449-1476, 2006.
- [2] M. Mutiso, and K. I. Winey, "Electrical properties of polymer nanocomposites containing Rod-Like nanofillers," *Progress in Polymer Sci.*, Vol. 40, pp. 63-84, 2015.
- [3] G. C. Montanari, D. Fabiani, F. Palmieri, D.K. Kaempfer, R. Thomann and R. Mulhaupt, "Modification of electrical properties and performance of EVA and PP insulation through nanostructure by organophilic silicates," *IEEE Trans. Dielectr. Electr. Insul.*, Vol. 11, pp. 754-762, 2004.
- [4] M. Ambid, D. Mary, G. Teyssedre, C. Laurent and G. C. Montanari, "Optical properties and luminescence behavior of PP/clay nanocomposites," *Trans. IEE Japan*, Vol. 126A, pp. 1-8, 2006.
- [5] C. T. Morse and G. J. Hill, "The electric strength of alumina: the effect of porosity," *Proc. Br. Ceram. Soc.*, Vol. 18, pp. 23-35, 1970.
- [6] <http://www.seprod.com/product/bulletins/PB%20Cloisite%2020A.pdf>.
- [7] L. A. Utracki, "Clay-Containing Polymeric Nanocomposites and Their Properties," *IEEE Electr. Insul. Mag.*, Vol. 26, No. 4, pp 6-15, 2010.
- [8] S. Fakhfakh, O. Jbara, M. Belhaj, Z. Fakhfakh, A. Kallel and E. I. Rau, "An experimental approach for dynamic investigation of the trapping properties of glass-ceramic under electron beam irradiation from a scanning electron microscope," *Europ. Phys. J. Appl.*, Vol. 21, pp. 137-146, 2003.
- [9] L. B. De Paiva, A. R. Morales and T. R. Guimaraes, "Structural and optical properties of polypropylene-montmorillonite nanocomposites," *Mater. Sci. Eng.*, Vol. 447, pp. 261-265, 2007.
- [10] L. Zhao, J. Li, S. Guo and Q. Du, "Ultrasonic oscillations induced morphology and property development of polypropylene/montmorillonite nanocomposites," *Polymer*, Vol. 47, pp. 2460-2469, 2006.
- [11] Q. Zheng, Y. Shangguan, S. Yan, Y. Song, M. Peng and Q. Zhang, "Structure, morphology and non-isothermal crystallization behavior of polypropylene catalloys," *Polymer*, Vol. 46, pp. 3163-3174, 2005.
- [12] P. H. Nam, P. Maitia, M. Okamoto, T. Kotaka, N. Hasegawa and A. Usuki, "A hierarchical structure and properties of intercalated polypropylene/clay nanocomposites," *Polymer*, Vol. 42, pp. 9633-9640, 2001.
- [13] M. S. Nazir, M. H. M. Kassim, L. Mohapatra, M. A. Gilani, M. R. Raza and K. Majeed, "Characteristic Properties of Nanoclays and Characterization of Nanoparticles and Nanocomposites," *Springer Science+Business Media Singapore*, M., *Engineering Materials*, DOI: 10.1007/978-981-10-1953-1_2, pp. 35-55, 2016.
- [14] Y. Di, S. Iannace, E. Di. Maio and L. Nicolais, "Nanocomposites by melt intercalation based on polycaprolactone and organoclay," *J. Polym. Sci.*, Vol. 41, pp. 670-678, 2013.
- [15] B. Lepoittevin, M. Devalckenaere, N. Pantoustier, M. Alexandre, D. Kubies D, C. Calberg and R. Jerome, "Poly(ϵ -caprolactone)/clay Dubois P nanocomposites prepared by melt intercalation: mechanical, thermal and rheological properties," *Polymer*, Vol. 43, pp. 4017-4023, 2002.
- [16] S. Labidi, N. Azema, D. Perrin and J. M. L. Cuesta, "Organo-modified montmorillonite/poly(ϵ -capro-lactone) nanocomposites prepared by melt intercalation in a twin-screw extruder," *Polym. Degrad. Stab.*, Vol. 95, pp. 382-388, 2010.
- [17] L. N. Luduena, J. M. Kenny, A. Vazquez and V. A. Alvarez, "Effect of clay organic modifier on the final performance of PCL/clay nanocomposites," *Mat. Sci. Eng. A*, Vol. 529, pp. 215-223, 2011.
- [18] Q. Yuan and R. D. K. Misra, "Impact fracture behavior of clay-reinforced polypropylene nanocomposites," *Polymer*, Vol. 47, pp. 4421-4433, 2006.
- [19] Y. Li, C. Han and J. Bian, "Crystallization and morphology studies of biodegradable poly(ϵ -caprolactone)/silica nanocomposites," *Polymer Compos. Vol.* 34, pp. 131-140, 2013.
- [20] A. Guevara-Morales and A. C. Taylor, "Mechanical and dielectric properties of epoxy-clay nanocomposites," *J. Mater. Sci.*, Vol. 49, pp. 1574-1584, 2014.
- [21] L. Dewimille, B. Bresson and L. Bokobza, "Synthesis, structure and morphology of poly (dimethylsiloxane) networks filled in situ generated silica particles," *Polymer*, Vol. 46, pp. 4135-4143, 2005.
- [22] S. K. O'Leary, S. R. Johnson and P. K. Lim, "The relationship between the distribution of electronic states and the optical absorption spectrum of an amorphous semiconductor: An empirical analysis," *J. Appl. Phys.*, Vol. 82, pp. 3334-3340, 1997.
- [23] F. S. Boydag, S. V. Mamedov, V. A. Alekperov and G. Kandemir, "A study of the optical properties of polypropylene based polymer composite films," *Macromol. Sympos.*, Vol. 146, pp. 187-192, 1999.
- [24] N. F. Mott and E. A. Davis, *Electronic Processes in Non-Crystalline Materials*, Clarendon Press, Oxford, UK, 1979.
- [25] S. John, C. Soukoulis, M. H. Cohen and E. N. Economou, "Theory of Electron Band Tails and the Urbach Optical-Absorption Edge," *Phys. Rev. Lett.*, Vol. 57, pp. 1777-1780, 1986.
- [26] A. Atta, A. M. Abdel Reheem and M. M. Abdel Rahman, "Surface treatment of polypropylene film using cold cathode ion source," *J. Nucl. Energy Sci. Generat. Technol.*, Vol. 4, pp. 1000129, 2015.
- [27] R. M. Radwan, "Electron induced modifications in the optical properties of polypropylene," *J. Phys. D: Appl. Phys.*, Vol. 40, pp. 374-379, 2007.
- [28] M. Magda, I. Khalil, G. M. Nas and M. Naeem, "Optical properties of modified grafted polypropylene," *J. Phys. D: Appl. Phys.*, Vol. 39, pp. 5305-5309, 2006.
- [29] S. A. Saqan, "Optical energy estimation of irradiated polypropylene," *Int'l. J. Polym. Mat. Polym. Biomat.*, Vol. 55, pp. 405-410, 2007.
- [30] R. Mishra, S. P. Tripathy, D. Sinha, K. K. Dwivedi, S. Ghosh, D. T. Khathing, M. Müller, D. Fink and W. H. Chung, "Optical and electrical properties of some electron and proton irradiated polymers," *Nucl. Instrum. Methods Phys. Res. Sect. B*, Vol. 168, pp. 59-64, 2000.
- [31] S. K. F. Ahmed, M. W. Moon, Ch. Kim, Y. Jang, S. Han, J. Y. O. Choi, W. W. Park and K. R. Lee, "Optical properties of surface modified polypropylene by plasma immersion ion implantation technique," *Appl. Phys. Lett.*, Vol. 97, pp. 081908, 2010.
- [32] F. S. Boydag, Y. Lenger Ozcanl and V.A. Alekperov, "Comparison of optical properties and mechanical/electrical degradation of some low density polyethylene and propylene-based polymer blends," *J. Appl. Spectroscopy*, Vol. 74, pp. 752-753, 2007.

- [33] J. Cazaux, "Charging in Scanning Electron Microscopy from Inside and Outside," *Scanning*, Vol. 26, pp. 181-203, 2004.
- [34] M. Belhaj, O. Jbara, M. N. Filippov, E.I. Rau and M.V. Andriyov, "Analysis of two methods of measurement of surface potential of insulators in SEM: electron spectroscopy and X-ray spectroscopy methods," *Appl. Surface. Sci.*, Vol. 177, pp. 58-65, 2001.
- [35] S. Fakhfakh, O. Jbara, S. Rondot, A. Hadjadj, J. M. Patat and Z. Fakhfakh, "Analysis of electrical charging and discharging kinetics of different glasses under electron irradiation in a scanning electron microscope," *J. Appl. Phys.*, Vol. 108, pp. 093705, 2010.
- [36] M. Abou-Dakka, "Evolution of Space Charges and Conductivity with DC Aging of Polyethylene-Synthetic and Natural Clay," *Composites. Nanomaterials*, Vol. 463748, pp. 1-8, 2012.
- [37] Y. Huang, K. Wu, M. Bell, A. Oakes, T. Ratcliff, N. A. Lanzillo, C. Breneman, B. C. Benicewicz and L. S. Schadler, "The Effects of Nanoparticles and Organic Additives with Controlled Dispersion on Dielectric Properties of Polymers: Charge Trapping and Impact Excitation," *J. Appl. Phys.*, Vol. 120, art. 055102, 2016.



Omar Jbara was born in Morocco in 1958. He received the degrees of M.Sc. in physics in 1982 and the Ph.D. in physics in 1990 at the Reims University, France. He is currently a Professor at the Physics department of the Reims University. Among his research interests are the development of new approaches in order to study insulating materials under electron irradiation in a scanning electron microscope.



Slim Fakhfakh was born in Tunisia in 1974. He received the degree of M.Sc. in physics in 2000 and the Ph.D. in physics in 2004 at the Reims University (France). He is currently a Lecturer at the Physics department of the Sfax University. Among his research interests are the development of new approaches in order to study electric properties of insulating materials submitted to an electronic irradiation in a scanning electron microscope.



Aomar Hadjadj was born in 1960 in Algeria. He received the rank of lecturer of the University of Oran (Algeria) in 1986 and the Ph.D. degree in materials physics in 1996 from the Ecole Polytechnique (France). He is currently a professor of physics at the University Institute of Technology of Reims - His research interest deals with the optoelectronic properties of semiconductor materials involved in the photovoltaic conversion.



Sébastien Rondot was born in France in July 1968. He received the degree of M.Sc. in 1990 from the University of Besançon, France and the Ph.D. in physics in 1995 from the University of Reims, France. He is now a Lecturer at the Physics department of Reims University. His research interest concerns study of electric properties of dielectric materials submitted to an electronic irradiation in a scanning electron microscope.



Latifa Ben Ammar was born in Tunisia in August 1989. She received the degree of M.Sc. in 2013 from the University of Sfax, Tunisia. She is now a Ph.D. student at the Physics department of Sfax University. Her research field is the characterization of nanocomposites materials.